Image Processing & Recognition: Lecture 4 Review

Welcome to Level 4! Previously, we looked at static images. Now, we add the dimension of **Time**. This lecture covers how machines "remember" the past to predict the future.

Part 1: What are you learning? (The "Human" Summary)

1. The Problem: Static vs. Sequential

- **Previous Models (MLP/CNN):** They have "amnesia." If you show them a video frame of a ball in the air, they don't know if it's going up or coming down because they don't remember the previous frame.
- **The Solution (RNN):** A network with a "Hidden State" (Memory). It takes an input (X_t) AND its own memory from the previous step (H_{t-1}) to decide what to do.

2. The Core Mechanism: The Loop

- **Input-to-Hidden:** We process the current data (like today's stock price).
- **Hidden-to-Hidden:** We add the memory of yesterday's context.
- **The Equation:** $H_t = \phi(X_t W_{xh} + H_{t-1} W_{hh} + b)$.
  - New Memory = Activation(Current Input info + Past Memory info).

3. Advanced RNNs (The "Fixes")

Basic RNNs struggle to remember things for a long time (the **Vanishing Gradient** problem).

- **GRU (Gated Recurrent Unit):** Has "gates" (Reset and Update) to decide what to forget and what to keep.
- **LSTM (Long Short-Term Memory):** The gold standard for years. It has a separate "Cell State" and three gates (Input, Forget, Output) to carefully manage long-term memory.

4. Sequence-to-Sequence (Seq2Seq)

- **Encoder:** Reads a sequence (like an English sentence) and compresses it into a "Context Vector."
- **Decoder:** Takes that vector and generates a new sequence (like a French sentence). This is how Google Translate works!

Part 2: The Solved Problem (Step-by-Step)

Let's decode **Exercise 4** from your PDF (Page 97).

The Task: Run a simple RNN forward for 2 time steps ($t = 1, t = 2$).

- **Setup:** Batch size $n = 2$ (2 samples processed at once).

- **Dimensions:** Input $d = 2$, Hidden $h = 3$, Output $q = 2$.
- **Activation:** ReLU ($\phi(x) = \max(0, x)$).

The Matrices (Given in Solution):

- **Inputs:** $X_1 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
- **Weights:**
  - W_{xh} (Input $\to$ Hidden): $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
  - W_{hh} (Hidden $\to$ Hidden): $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$
  - W_{hq} (Hidden $\to$ Output): $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$
- **Biases:** $b_h = [1, 0, -1]$, $b_q = [1, -1]$.
- **Initial Memory:** $H_0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (All zeros).

Step 1: Time Step 1 ($t = 1$)

Goal: Calculate New Memory H_1 and Output O_1.

1. **Calculate Input Term ($X_1 W_{xh}$):**

$$\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

2. **Calculate Memory Term ($H_0 W_{hh}$):** Since H_0 is all zeros, this term is all zeros.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

3. **Add Bias (b_h) & Apply ReLU:** Add $[1, 0, -1]$ to every row of the Input Term.

$$\text{Sum} = \begin{bmatrix} 1+1 & 0+0 & 1-1 \\ 1+1 & -1+0 & 1-1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \end{bmatrix}$$

Apply ReLU (set negatives to 0):

$$H_1 = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

4. **Calculate Output ($O_1 = H_1 W_{hq} + b_q$):**

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}$$

Add Bias $[1, -1]$:

$$O_1 = \begin{bmatrix} 2+1 & 0-1 \\ 2+1 & 0-1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix}$$

Step 2: Time Step 2 ($t = 2$)

Goal: Calculate New Memory H_2 and Output O_2. *Crucial:* We use H_1 (calculated above) as the memory input!

1. **Calculate Input Term ($X_2 W_{xh}$):**

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

2. **Calculate Memory Term ($H_1 W_{hh}$):**

$$\begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -2 \\ 2 & 0 & -2 \end{bmatrix}$$

3. **Add Bias (b_h) & Combine:**

$$\text{Input Term} + \text{Memory Term} + \text{Bias}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 & -2 \\ 2 & 0 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & -3 \\ 4 & 0 & -2 \end{bmatrix}$$

Apply ReLU:

$$H_2 = \begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

4. **Calculate Output ($O_2 = H_2 W_{hq} + b_q$):**

$$\begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 4 & 0 \end{bmatrix}$$

Add Bias $[1, -1]$:

$$O_2 = \begin{bmatrix} 4 & 0 \\ 5 & -1 \end{bmatrix}$$

Step 3: Parameter Counting

- **Input-to-Hidden (W_{xh}):** $2 \text{ (in)} \times 3 \text{ (hidden)} = 6.$
- **Hidden-to-Hidden (W_{hh}):** $3 \text{ (hidden)} \times 3 \text{ (hidden)} = 9.$
- **Hidden Bias (b_h):** $3.$
- **Hidden-to-Output (W_{hq}):** $3 \text{ (hidden)} \times 2 \text{ (out)} = 6.$
- **Output Bias (b_q):** $2.$
- **Total:** $6 + 9 + 3 + 6 + 2 = 26.$

Part 3: Your Turn (Interactive Exercises)

Practice Problem 1: The "Lite" RNN

Let's simplify the numbers so you can practice the flow.

Setup:

- **Input:** 1 sample, 2 dims. Sequence length 2.
  - $X_1 = [1, 0]$
  - $X_2 = [0, 1]$
- **Memory:** Size 2. $H_0 = [0, 0]$.
- **Activation:** ReLU.
- **Weights:**
  - $W_{xh} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (Input adds to memory)
  - $W_{hh} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (Identity matrix - keeps memory exactly as is)
  - Bias $b_h = [0, 0]$

Task: Calculate the Hidden State at time step 2 (H_2).

Solving Explanation:

1. **Step 1 ($t = 1$):**

- Input: $X_1 \times W_{xh} = [1, 0] \times \text{ones} = [1, 1]$.
- Memory: $H_0 \times W_{hh} = [0, 0]$.
- $H_1 = \text{ReLU}([1, 1] + [0, 0]) = [1, 1]$.

2. **Step 2 ($t = 2$):**
   - Input: $X_2 \times W_{xh} = [0, 1] \times \text{ones} = [1, 1]$.
   - Memory: **Crucial Step** -> We use H_1. $H_1 \times W_{hh} = [1, 1] \times \text{Identity} = [1, 1]$.
   - Combine: Input Term + Memory Term = $[1, 1] + [1, 1] = [2, 2]$.
   - $H_2 = \text{ReLU}([2, 2]) = [2, 2]$.

Answer: $H_2 = [2, 2]$. (The network accumulated the inputs!)

Practice Problem 2: Parameter Counting (Exam Style)

Setup: You are building a Chatbot using a standard RNN.

- **Input:** Word vectors of size 100 ($d = 100$).
- **Hidden State:** Size 50 ($h = 50$).
- **Output:** Vocabulary size 1000 ($q = 1000$).

Task: Calculate the number of trainable parameters in the **Hidden Layer** only (Input-to-Hidden + Hidden-to-Hidden + Bias).

Solving Explanation:

1. **Input-to-Hidden Matrix (W_{xh}):** Size = $100 \text{ (input)} \times 50 \text{ (hidden)} = 5,000$.
2. **Hidden-to-Hidden Matrix (W_{hh}):** Size = $50 \text{ (hidden)} \times 50 \text{ (hidden)} = 2,500$.
3. **Bias (b_h):** Size = 50 (hidden).